EMNLP BoF 2026 · October 27 · 9:00–10:30 AM

Let's talk about:
Symbolic and Trustworthy Reasoning Across Language and Vision

Panel · Lightning talks · Open discussion

Organized by Shreya Rajpal, Michigan State University

EMNLP2026 BUDAPEST

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How and when can structured and symbolic methods make reasoning more verifiable and trustworthy?

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Symbolic reasoning and trustworthiness

Symbolic reasoning

Knowledge is encoded as explicit symbolic structures (entities, relations, rules). Reasoning applies rules to these structures to derive new ones.

`left(A,B) ∧ left(B,C)`
`⇒ left(A,C)`

rule: transitivity

Trustworthiness

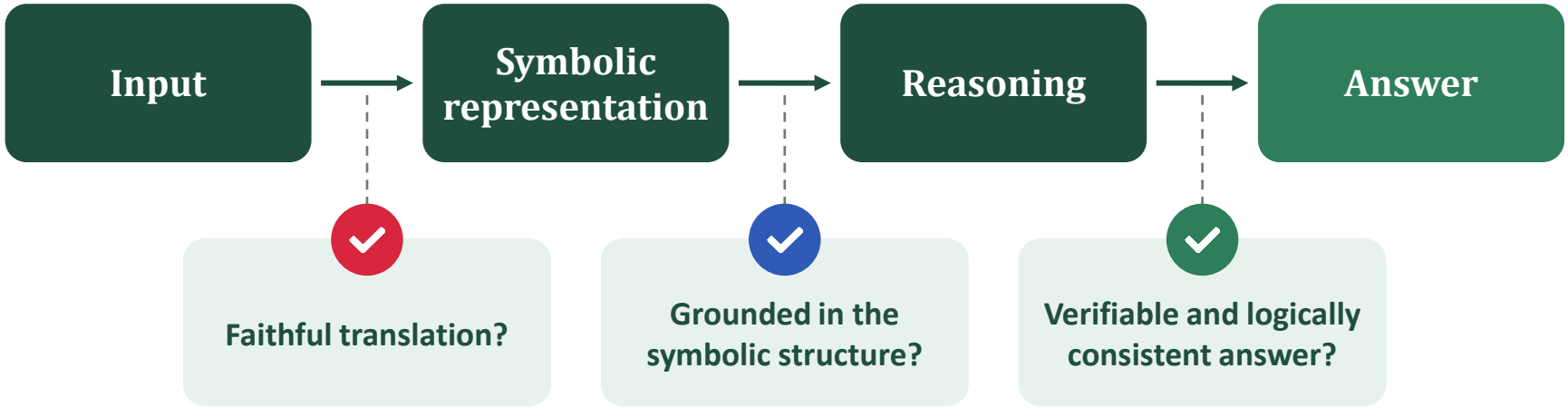
A model's answer is trustworthy when it's reasoning is based on the right evidence and is logically coherent.

Faithful: reflects how the model actually reached the answer

Plausible: is coherent and convincing to a human reader

Sources: Newell & Simon (1976), Computer Science as Empirical Inquiry: Symbols and Search, CACM · Jacovi & Goldberg (2020), Towards Faithfully Interpretable NLP Systems, ACL · Shailya, Rajpal, Krishnan & Ravindran (2025), LExT: Towards Evaluating Trustworthiness of Natural Language Explanations, FAccT

Symbolic structure for trustworthy reasoning



Explicit symbols make each step inspectable, so faithfulness can be checked at every step, not only at the final answer.

Quick show of hands



You build structured or symbolic reasoning systems



You work on making model reasoning trustworthy



You use structure to make reasoning trustworthy

Send us your questions

Scan to submit a question for the panel or the speakers

We'll pick up submitted questions during the open discussion.



Today's program



9:00
Welcome & panel introductions

9:06
Opening lightning talk

9:15
Three lightning talks

9:45
Panel discussion

10:15
Audience Q&A & open discussion

Ends 10:30 · Questions welcome throughout

Our panel



Daniel Preotiuc-Pietro

Bloomberg

Structured reasoning & calibration



Leslie Barrett

Bloomberg Law

Faithful & structured reasoning



Parisa Kordjamshidi

Michigan State University

Language, vision & symbolic reasoning



Artem Shelmanov

MBZUAI

Uncertainty & reliable LLMs

Lightning talks

1

LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning

Qingjing Chen

PhD candidate, University of Bologna
EMNLP Findings 2026

2

Transforming Interpretability Gains to Explainability Wins

Gokul S Krishnan

Senior Research Director, Centre for Responsible AI, IIT Madras

3

Grammar of the Wave: Towards Explainable Multivariate Time Series Event Detection via Neuro-Symbolic VLM Agents

Sky Chenwei Wan

PhD Student, SLB, France
EMNLP 2026 (Main)

5 min talk + 5 min discussion each

Opening lightning talk

When does structure make reasoning trustworthy?

Shreya Rajpal · Michigan State University

1 Reasoning in natural language

2 Symbolic translation

3 Results

4 Other applications

Lightning talk 1 of 3



Transforming Interpretability Gains to Explainability Wins

Gokul S Krishnan

Senior Research Director, Centre for Responsible AI, IIT Madras

5 min talk + 5 min discussion

Lightning talk 2 of 3

LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning

Qingjing Chen
PhD candidate, University of Bologna
EMNLP Findings 2026

5 min talk + 5 min discussion

Lightning talk 3 of 3

Grammar of the Wave: Towards Explainable Multivariate Time Series Event Detection via Neuro-Symbolic VLM Agents

Sky Chenwei Wan
PhD Student, SLB, France
EMNLP 2026

5 min talk + 5 min discussion

Panel discussion

1. How can neuro-symbolic AI make reasoning more trustworthy?
2. When do rules and programs help, and how should they pair with foundation models?
3. Beyond accuracy, what evidence shows that reasoning is trustworthy?
4. Where should neuro-symbolic and trustworthy reasoning go next?

Daniel Preotiuc-Pietro · Leslie Barrett · Parisa Kordjamshidi · Artem Shelmanov

Thank you

Questions and open discussion

Panel: Daniel Preotiuc-Pietro · Leslie Barrett · Parisa Kordjamshidi · Artem Shelmanov

Lightning talks: Qingjing Chen · Gokul S Krishnan · Sky Chenwei Wan

Organizer: Shreya Rajpal, Michigan State University

